

SUPERSINGULAR PRIMES COMMON TO TWO ELLIPTIC CURVES

Étienne FOUVRY (Orsay) and M. Ram MURTY (Montreal)

I. INTRODUCTION

Let E be an elliptic curve over $\mathbf{Q}$. Denote by $\pi_0(x, E)$ the cardinality of the set of the supersingular primes of E less than x . It has been conjectured by Lang and Trotter [L–T] that, when E has no complex multiplication, the following holds, when $x \rightarrow \infty$

$$(1.1) \quad \pi_0(x, E) \sim C_E \frac{\sqrt{x}}{\log x}$$

where C_E is a positive constant depending only on E , precisely defined in terms of $\text{Gal}(\mathbf{Q}(E_{\text{tors}}), \mathbf{Q})$.

The first significant step towards (1.1) is due to Elkies ([El1]), who proved that each elliptic curve over $\mathbf{Q}$, has infinitely many supersingular primes. This result was improved by the authors, who proved

THEOREM A ([F–M] Théorème 1) *Let E be an elliptic curve over $\mathbf{Q}$. Then, for every positive δ , there exists $x_0(\delta, E)$ such that, for $x > x_0(\delta, E)$, the following holds:*

$$\pi_0(x, E) > \frac{\log_3 x}{(\log_4 x)^{1+\delta}}.$$

Here $\log_k$ is the k -fold iterated logarithm function. Note that the best upperbound for $\pi_0(x, E)$ is due to Elkies and Murty ([El2], [El3]) and has the shape $\pi_0(x, E) = O_E(x^{\frac{3}{4}})$, for any non CM-curve E , with the convention that CM means that the curve has complex multiplication.

In the direction of (1.1), we must also quote another result which asserts, very vaguely speaking, that the Lang–Trotter Conjecture is true *on average*. More precisely, let a, b be two integers with $4a^3 + 27b^2 \neq 0$ and let $E_{a,b}$ be the elliptic curve defined by the equation

$$y^2 = x^3 + ax + b,$$

then we have

THEOREM B ([F–M] Théorème 6) *For every positive ε , we have, for $x \rightarrow \infty$, the asymptotic relation*

$$\sum_{|a| \leq A} \sum_{|b| \leq B} \pi_0(x, E_{a,b}) \sim \left(\frac{\pi}{3} \cdot \frac{\sqrt{x}}{\log x} \right) \cdot (4AB)$$

uniformly for $A \geq x^{\frac{1}{2}+\varepsilon}$, $B \geq x^{\frac{1}{2}+\varepsilon}$, $AB \geq x^{\frac{3}{2}+\varepsilon}$.

A familiar way to write (1.1) is to say that the probability for a prime p to be supersingular for E is

$$(1.2) \quad \frac{C_E}{2} \cdot \frac{1}{\sqrt{p}},$$

and we are led to the problem of the primes supersingular for two given elliptic curves E and E' . We say that two elliptic curves over $\mathbf{Q}$ are *in general position* when none of them has complex multiplication and when they are not isogenous over $\bar{\mathbf{Q}}$. Let us recall that, when E has complex multiplication, we have $\pi_0(x, E) \sim \frac{x}{2 \log x}$, and if E and E' are isogenous over $\bar{\mathbf{Q}}$, they have the same supersingular primes apart from the prime divisors of the conductors of these curves; in other words, we have

$$(1.3) \quad \pi_0(x, E, E') = \pi_0(x, E) - O_E(1) = \pi_0(x, E') - O_{E'}(1),$$

where $\pi_0(x, E, E')$ is the cardinality of the set of primes, less than x , supersingular for E and E' .

Then, following (1.2), it is natural to think that if E and E' are in general position, the probability for a prime p to be supersingular for E and E' should be

$$\frac{C_{E,E'}}{p},$$

where $C_{E,E'}$ is a positive constant depending only on E and E' .

Such a probabilistic assumption of independence appears in [L–T] page 37, and leads to the conjecture

$$(1.4) \quad \pi_0(x, E, E') \sim C_{E,E'} \log_2 x \quad (x \rightarrow \infty).$$

when E and E' are supposed to be in general position. This conjecture seems extremely hard to prove – for the moment, nobody knows how to prove that $\pi_0(x, E, E') \rightarrow \infty$ when $x \rightarrow \infty$ – since the set of primes in question is very, very sparse (heuristically as sparse as the following set connected with the Mersenne conjecture: $\{p \leq x; 2^p - 1 \text{ is a prime}\}$).

The bulk of this paper is to prove that (1.4) is true *on average*, in the same philosophy as Theorem B proves (1.1) on average. We will prove

THEOREM 1 *For every positive ε , we have for $x \rightarrow \infty$, the asymptotic relation*

$$(1.5) \quad \sum_{|a| \leq A} \sum_{|a'| \leq A'} \sum_{|b| \leq B} \sum_{|b'| \leq B'} \pi_0(x, E_{a,b}, E_{a',b'}) \sim \frac{35}{96} \cdot \log_2 x \cdot (16AA'BB')$$

holds uniformly for $A, A' \geq x^{\frac{1}{2}+\varepsilon}$, $B, B' \geq x^{\frac{1}{2}+\varepsilon}$, $AB, A'B' \geq x^{\frac{3}{2}+\varepsilon}$.

Let $S(A, A', B, B')$ be the sum studied in (1.5). To be allowed to say that Theorem 1 proves (1.4) on average, we must check that the contribution to $S(A, A', B, B')$ of the

pairs of curves $(E_{a,b}, E_{a',b'})$ which are not in general position is negligible. So we denote respectively by $S^{\text{CM}}(A, A', B, B')$ and $S^{\text{iso}}(A, A', B, B')$ the contribution of those pairs with $E_{a,b}$ having complex multiplication and with $E_{a,b}$ and $E_{a',b'}$ isogenous over $\bar{\mathbf{Q}}$.

The first contribution satisfies :

$$S^{\text{CM}}(A, A', B, B') \leq |\{(a, b); |a| \leq A, |b| \leq B, E_{a,b} \text{ is CM}\}| \cdot \sum_{|a'| \leq A'} \sum_{|b'| \leq B'} \pi_0(x, E_{a',b'}).$$

There are thirteen families of elliptic curves with complex multiplication, they can be written as

$$E_{0,t}; E_{t,0}; E_{\alpha_i t^2, \beta_i t^3} \quad (t \in \mathbf{Z}^*, 1 \leq i \leq 11)$$

where the (α_i, β_i) are eleven pairs of integers. Theorem B implies under the conditions of Theorem 1

$$S^{\text{CM}}(A, A', B, B') = O\left(\max(A, B) \cdot A' B' \frac{\sqrt{x}}{\log x}\right) = O(AA'BB'),$$

which is clearly negligible, compared to the expected main term.

The term $S^{\text{iso}}(A, A', B, B')$ requires a more delicate treatment. We shall prove later the following :

LEMMA 1 *Let E be an elliptic curve over $\mathbf{Q}$. Then , for A and B tending to infinity, we have*

$$\begin{aligned} & |\{(a, b); 0 < |a| \leq A, 0 < |b| \leq B, E_{a,b} \text{ isogenous over } \bar{\mathbf{Q}} \text{ to } E\}| \\ &= O(\min(A^{\frac{1}{2}}, B^{\frac{1}{3}}) \log^9(2AB)). \end{aligned}$$

This lemma and Theorem B directly imply

$$\begin{aligned} & S^{\text{iso}}(A, A', B, B') \\ & \leq \max_{E/\mathbf{Q}} |\{(a, b); |a| \leq A, |b| \leq B, E_{a,b} \text{ isogenous over } \bar{\mathbf{Q}} \text{ to } E\}| \cdot \sum_{|a'| \leq A'} \sum_{|b'| \leq B'} \pi_0(x, E_{a',b'}) \\ &= O(\min(A^{\frac{1}{2}}, B^{\frac{1}{3}}) \cdot A' B' \sqrt{x} \log^{10}(2AB)) = O(AA'BB') \end{aligned}$$

which is also negligible.

Proof of Lemma 1.

Since $E_{a,b}$ and E are isogenous over $\bar{\mathbf{Q}}$, the isogeny is defined on a number field K of relative degree at most 12 over $\mathbf{Q}$ ([M–W] Lemma 6.1 for instance). Now, we use a strong theorem of Masser and Wustholz ([M–W] Theorem), asserting, that if there is an isogeny between two elliptic curves E and E' over a number field k of degree at most d over $\mathbf{Q}$, then there exists between them a “ simple ” isogeny, i.e. with a degree less than $c(w(E'))^4$, where c depends only on d and $w(E')$ is the maximum of 1 and of the logarithmic Weil height of the curve E' . In our situation, we deduce that, if E and $E_{a,b}$

are isogenous, there is an isogeny of degree $O(\log^4(2AB))$. A simple trick of algebra ([M–W] Lemma 6.2) allows us to suppose that this isogeny is cyclic.

Let j_E be the invariant of E . The modular polynomial of order n , $\Phi_n(X, j_E)$ ([La] pages 55–59) detects the existence of a cyclic isogeny of degree n between $E_{a,b}$ and E . More precisely, such an isogeny exists if and only if we have

$$\Phi_n\left(\frac{6912a^3}{4a^3 + 27b^2}, j_E\right) = 0.$$

The degree in X of $\Phi_n(X, j_E)$ is $n \prod_{p|n} (1 + \frac{1}{p}) = O(n \log 2n)$, which implies that the equation $\Phi_n(X, j_E) = 0$ has at most $O(n \log 2n)$ solutions in $\mathbf{Q}$. Let $\frac{u}{v}$ be such a root, which we can suppose different from 0 and 1728 since $ab \neq 0$. The equation in a and b

$$\frac{6912a^3}{4a^3 + 27b^2} = \frac{u}{v}$$

has $O(\min(A^{\frac{1}{2}}, B^{\frac{1}{3}}))$ roots in the rectangle $[-A, A] \times [-B, B]$. Then gathering these solutions when $\frac{u}{v}$ varies and when n varies, we complete the proof of Lemma 1.

The above discussion gives another formulation of Theorem 1:

THEOREM 2 *Let $\dagger$ indicates that $E_{a,b}$ and $E_{a',b'}$ are in general position. Then under the conditions of Theorem 1, we have*

$$(1.6) \quad \sum_{|a| \leq A} \sum_{|a'| \leq A'} \sum_{|b| \leq B} \sum_{|b'| \leq B'}^{\dagger} \pi_0(x, E_{a,b}, E_{a',b'}) \sim \left(\frac{35}{96} \cdot \log_2 x\right) \sum_{|a| \leq A} \sum_{|a'| \leq A'} \sum_{|b| \leq B} \sum_{|b'| \leq B'}^{\dagger} 1.$$

A more difficult question seems to give an asymptotic formula similar to (1.6), but where $E_{a,b}$ and $E_{a',b'}$ takes only one value in each isogeny class.

II. FROM SUPERSINGULAR PRIMES TO CLASS NUMBERS.

The starting point of our proof is

LEMMA 2 *Let $p \geq 5$ be a prime. The number of isomorphism classes of elliptic curves over $\mathbf{F}_p$ with $p+1$ points is equal to $H(-4p)$ (number of of isomorphism classes of positive quadratic forms, not necessary primitive, with discriminant $-4p$).*

Such a statement appears at several places in the literature, for instance in [Bi], page 58. Since the quadratic forms counted by $H(-4p)$ are not necessarily primitive, we have the equality

$$(2.1) \quad H(-4p) = h(-p) + h(-4p)$$

with the convention that the h symbol is equal to zero when it is not defined. We denote by $y^2 = x^3 + a_i x + b_i$ ($1 \leq i \leq H(-4p)$) the equations which define representatives of the above isomorphism classes of elliptic curves over $\mathbf{F}_p$. We can suppose that these equations are minimal relative to p , for instance by imposing the conditions $0 \leq a_i < p$, $0 \leq b_i < p$. If the equation defining $E_{a,b}$ is minimal for p we deduce that p is supersingular for $E_{a,b}$ if and only if there exists $t \in \mathbf{F}_p^*$ and $1 \leq i \leq H(-4p)$, such that

$$(2.2). \quad a \equiv a_i t^4, \quad b \equiv b_i t^6 \pmod{p}$$

For $a_i b_i \not\equiv 0 \pmod{p}$, the image of the application $t \in \mathbf{F}_p^* \rightarrow (a_i t^4, b_i t^6)$ has cardinality $\frac{p-1}{2}$ for $p \geq 5$. We gather the above observations into

LEMMA 3 *Let $p \geq 5$, then there exists a set $\mathcal{E}_p$ of residue classes mod $p \times \text{mod } p$ with the following properties*

- i) $|\mathcal{E}_p| = \frac{H(-4p)p}{2} + O(p)$
- ii) if $(a, b) \in \mathbf{Z}^2 - \{(0, 0)\}$ and if k is the largest integer such that $p^{4k} | a$ and $p^{6k} | b$, then p is a supersingular prime for $E_{a,b}$, if and only if $(ap^{-4k}, bp^{-6k}) \pmod{p}$ belongs to $\mathcal{E}_p$.

It is now easy to transform $S(A, B, A', B')$ into

$$\begin{aligned} S(A, B, A', B') = & \sum_{p \leq x} \left(\sum_{(u,v) \in \mathcal{E}_p} \sum_{\substack{a \equiv u \pmod{p} \\ |a| \leq A}} \sum_{\substack{b \equiv v \pmod{p} \\ |b| \leq B}} 1 + O\left(\left(\frac{A}{p^4} + 1\right)\left(\frac{B}{p^6} + 1\right)\right) \right) \times \\ & \left(\sum_{(u,v) \in \mathcal{E}_p} \sum_{\substack{a' \equiv u \pmod{p} \\ |a'| \leq A'}} \sum_{\substack{b' \equiv v \pmod{p} \\ |b'| \leq B'}} 1 + O\left(\left(\frac{A'}{p^4} + 1\right)\left(\frac{B'}{p^6} + 1\right)\right) \right) + O(ABA'B'), \end{aligned}$$

where the errors terms come from the non-minimal equations and from the primes 2 and 3. Using the trivial equality

$$\sum_{\substack{a \equiv u \pmod{p} \\ |a| \leq A}} 1 = \frac{2A}{p} + O(1),$$

we get the equality

$$\begin{aligned} S(A, B, A', B') = & \sum_{p \leq x} |\mathcal{E}_p|^2 \frac{2A}{p} \cdot \frac{2B}{p} \cdot \frac{2A'}{p} \cdot \frac{2B'}{p} + \\ & + O\left(x^4 \log^2 x + (A + B + A' + B')x^3 \log^2 x + \right. \\ (2.3) \quad & \left. + (AB + \dots + A'B')x^2 \log^2 x + (ABA' + \dots + BA'B')x \log^2 x + ABA'B'\right), \end{aligned}$$

by using Lemma 3 *ii*) and by using the classical upperbound $h(-d) = O(\sqrt{d} \log d)$ (note that some log-factors could be spared by using average bounds of that quantity by methods of the next Proposition). We postpone to paragraph IV an improvement of (2.3) by appealing to the theory of exponential sums. The equation *i*) of Lemma 3 splits (2.3) into main term and error term:

$$(2.4) \quad S = MT + ET$$

with

$$MT(A, B, A', B') = 4AA'BB' \sum_{p \leq x} \frac{H^2(-4p)}{p^2}$$

and

$$(2.5) \quad ET(A, B, A', B') = O(ABA'B')$$

under the conditions

$$(2.6) \quad A, A', B, B' \geq x \log x.$$

III. CLASS NUMBERS ON AVERAGE.

To evaluate the main term in (2.4), we will prove the

PROPOSITION *For $x \rightarrow \infty$ we have*

$$\sum_{p \leq x} \frac{H^2(-4p)}{p^2} \sim \frac{35}{24} \log_2 x.$$

Such a formula, one more time illustrates the well known fact that on average $h(-d)$ behaves like $\sqrt{d}$ and we will treat it by the values of L -functions at the point 1. Formula (2.1) transforms our sum into

$$(3.1) \quad \sum_{p \leq x} \frac{h^2(-p)}{p^2} + 2 \sum_{p \leq x} \frac{h(-p)h(-4p)}{p^2} + \sum_{p \leq x} \frac{h^2(-4p)}{p^2} = T_{1,1}(x) + 2T_{1,4}(x) + T_{4,4}(x)$$

say. We will concentrate on the typical sum $T_{1,1}(x)$, for which we will prove

$$(3.2) \quad T_{1,1}(x) \sim \frac{5}{24} \log_2 x$$

but the proof of this formula can be obtained, after integrating by parts, from the following

$$(3.3) \quad T_{1,1}^*(x) := \sum_{p \leq x} \frac{h^2(-p)}{p} \sim \frac{5}{24} \frac{x}{\log x}.$$

The classical Dirichlet class number formula, coupled with the Polya–Vinogradov formula gives an expression of $h(-p)$ as a finite sum

$$h(-p) = \frac{\sqrt{p}}{\pi} \sum_{n \leq U} \frac{\chi_{-p}(n)}{n} + O\left(\frac{p \log p}{U}\right),$$

for any $U > 1$ and χ_{-p} , the Kronecker symbol associated to $-p$ for $p \equiv 3 \pmod{4}$. Squaring this equality, we obtain for $\sqrt{x} \leq U \leq x$

$$(3.4) \quad T_{1,1}^*(x) = \frac{1}{\pi^2} \sum_{p \leq x} \sum_{n_1 \leq U} \sum_{n_2 \leq U} \frac{\chi_{-p}(n_1 n_2)}{n_1 n_2} + O\left(\frac{x^{\frac{3}{2}} \log x}{U}\right).$$

To treat the triple sum in (3.4), we put $n_1 n_2 = m l^2$ with m squarefree and $d_U(n)$ the modified divisor function

$$d_U(n) := |\{(n_1, n_2); n_1 \leq U, n_2 \leq U, n_1 n_2 = n\}|,$$

the main term in (3.4) is now

$$\frac{1}{\pi^2} \sum_m \frac{\mu^2(m)}{m} \sum_{p \leq x} \sum_l \frac{\chi_{-p}(m) \chi_{-p}^2(l)}{l^2} d_U(m l^2).$$

The most important contribution comes from $m = 1$, it has the shape

$$\frac{1}{\pi^2} \sum_{l \leq U} \frac{d_U(l^2)}{l^2} \left(\sum_{\substack{p \leq x, p \nmid l \\ p \equiv 3 \pmod{4}}} 1 \right) = \left(\frac{1}{2\pi^2} + o(1) \right) \cdot \frac{x}{\log x} \cdot \left(\sum_{l=1}^{\infty} \frac{d(l^2)}{l^2} + O(U^{-\frac{1}{4}}) \right).$$

But we have the equality

$$\sum_{l=1}^{\infty} \frac{d(l^2)}{l^2} = \prod_p \sum_{k=0}^{\infty} \frac{2k+1}{p^{2k}} = \prod_p \frac{1+1/p^2}{(1-1/p^2)^2} = \frac{\zeta^3(2)}{\zeta(4)} = \frac{5\pi^2}{12}.$$

Thus (3.4) may be written as

$$(3.5) \quad \frac{5}{24} \frac{x}{\log x} (1 + o(1)) + O\left(\sum_l \frac{1}{l^2} \left| \sum_{p \leq x} \sum_{m \neq 1} \frac{\mu^2(m) \chi_{-p}(m) \chi_{-p}^2(l)}{m} d_U(m l^2) \right| \right).$$

Since $m \neq 1$, we hope cancellation from summation over p and m of the terms $\chi_{-p}(m)$. We call $W(x, l)$ the double sum over these variables in the error term of (3.5). After cutting the ranges of summation, we have :

$$W(x, l) = O(\log^2 x \{ \sup_{\substack{M > 1 \\ 1 \leq P \leq x}} |W(l, M, P)| + x^\varepsilon \})$$

with

$$W(l, M, P) = \sum_{P < p \leq 2P} \sum_{M < m \leq 2M} \frac{\mu^2(m) \chi_{-p}(m)}{m} d_U(ml^2)$$

and the error term x^ε coming from the p dividing l . In $W(l, M, P)$, m is never divisible by 4, so we put $m' = m$ or $m' = m/2$ if m is odd or even; if we fix the congruence of p and m' modulo 8, we see that the Kronecker symbol is expressed in terms of the Jacobi symbol

$$\chi_{-p}(m) = \left(\frac{m'}{p}\right) \cdot \varepsilon_1 = \left(\frac{p}{m'}\right) \cdot \varepsilon_2$$

where ε_1 and ε_2 of absolute value 1, are independent of m and p . We now appeal to a general upperbound for a double sum over Jacobi symbols:

LEMMA 4 *Let (a_m) and (b_n) be complex numbers. Then we have*

$$\sum_{M < m \leq 2M} \sum_{N < n \leq 2N} a_m b_n \left(\frac{m}{n}\right) \ll \|a\|_2 \|b\|_2 (M^{\frac{1}{2}} + M^{\frac{1}{4}} N^{\frac{23}{32}}),$$

where the sum is over odd square free integers m and n .

Remark that the reciprocity law for Jacobi symbols implies that, in the above sum the variables m and n play a similar rôle and the following upperbound will be sufficient for our purpose:

$$(3.6) \quad \sum_{M < m \leq 2M} \sum_{N < n \leq 2N} a_m b_n \left(\frac{m}{n}\right) \ll \|a\|_2 \|b\|_2 (MN)^{\frac{1}{2}} (\min(M, N))^{-\delta},$$

for some absolute positive δ .

Our proof is quite standard and mixes Cauchy–Schwarz inequality and Burgess bound for character sum ([Bu] Theorem 2 with $\varepsilon = 1/32$) . This proof is a slight generalisation of [H–B], Lemma 4. We have

$$\begin{aligned} \left| \sum_m a_m \sum_n b_n \left(\frac{m}{n}\right) \right| &\leq \|a\|_2 \left\{ \sum_m \left| \sum_n b_n \left(\frac{m}{n}\right) \right|^2 \right\}^{\frac{1}{2}} = \|a\|_2 \left\{ \sum_{n_1} \sum_{n_2} b_{n_1} \bar{b}_{n_2} \sum_m \left(\frac{m}{n_1 n_2}\right) \right\}^{\frac{1}{2}} \\ &\ll \|a\|_2 (\|b\|_2^2 M + \sum_{n_1} \sum_{n_2} |b_{n_1} b_{n_2}| M^{\frac{1}{2}} N^{\frac{7}{16}})^{\frac{1}{2}} \ll \|a\|_2 \|b\|_2 M^{\frac{1}{2}} + \|a\|_2 \|b\|_2 M^{\frac{1}{4}} N^{\frac{23}{32}}. \end{aligned}$$

Note that the classical Polya–Vinogradov would be sufficient to prove (3.6), if the situation $M^{1-\delta_1} < N < M^{1+\delta_1}$ (with δ_1 a very little positive constant) was excluded, Burgess

bound is used to cover that case. This bound, which depends on a result of Weil, could be replaced by the much more accessible (Corollary of [F–I]), if that last result was stated for general moduli, not necessarily prime moduli.

We now use (3.6) to bound $W(l, M, P)$:

$$(3.7) \quad W(l, M, P) \ll M^{-1} \left(\sum_m d_U^2(ml^2) \right)^{\frac{1}{2}} M^{\frac{1}{2}} P(\min(M, P))^{-\delta} \\ \ll d(l^2) P \log^{\frac{3}{2}} x (\min(M, P))^{-\delta} \ll l^{\frac{1}{2}} x \log^{-10} x$$

as soon as we have $M \geq (\log x)^{\frac{100}{\delta}}$. In the case where $M \leq (\log x)^{\frac{100}{\delta}}$, (3.7) is a direct application of the famous Siegel–Walfisz Theorem on the distribution of primes in arithmetic progressions. Inserting this bound in (3.5), summing over l , we obtain (3.3) and by the way (3.2).

The other terms $T_{1,4}(x)$ and $T_{4,4}(x)$ are evaluated along the same techniques. We write

$$T_{1,4}^*(x) = \frac{2}{\pi^2} \sum_{p \leq x} \sum_{n_1 \leq U} \sum_{n_2 \leq U} \frac{\chi_{-p}(n_1) \chi_{-4p}(n_2)}{n_1 n_2} + O\left(\frac{x^{\frac{3}{2}} \log x}{U}\right).$$

In the above sum, we may suppose that n_2 is odd, this implies that $\chi_{-4p}(n_2) = \chi_{-p}(n_2)$. If we write

$$d_U^*(n) = |\{(n_1, n_2); n_1 \leq U, n_2 \leq U, n_1 n_2 = n, n_2 \text{ odd}\}|,$$

we obtain the equality

$$(3.8) \quad T_{1,4}^*(x) \sim \frac{2}{\pi^2} \sum_{l \leq U} \frac{d_U^*(l^2)}{l^2} \left(\sum_{\substack{p \leq x, p \nmid l \\ p \equiv 3 \pmod{4}}} 1 \right) \\ \sim \frac{2}{\pi^2} \cdot \frac{4}{3} \cdot \prod_{p \geq 3} \frac{1 + 1/p^2}{(1 - 1/p^2)^2} \cdot \frac{x}{2 \log x} \sim \frac{1}{4} \cdot \frac{x}{\log x}$$

Similarly, we have the equality

$$T_{4,4}^*(x) = \frac{4}{\pi^2} \sum_{p \leq x} \sum_{n_1 \leq U} \sum_{n_2 \leq U} \frac{\chi_{-4p}(n_1) \chi_{-4p}(n_2)}{n_1 n_2} + O\left(\frac{x^{\frac{3}{2}} \log^2 x}{U}\right)$$

and we may suppose that both n_1 and n_2 are odd. We define

$$d_U^{**}(n) = |\{(n_1, n_2); n_1 \leq U, n_2 \leq U, n_1 n_2 = n, n_1 \text{ and } n_2 \text{ are odd}\}|,$$

and we obtain

$$\begin{aligned}
T_{4,4}^*(x) &\sim \frac{4}{\pi^2} \sum_{l \leq U} \frac{d_U^{**}(l^2)}{l^2} \left(\sum_{p \leq x, p \nmid l} 1 \right) \\
(3.9) \quad &\sim \frac{4}{\pi^2} \prod_{p \geq 3} \frac{1 + 1/p^2}{(1 - 1/p^2)^2} \cdot \frac{x}{\log x} \sim \frac{3}{4} \cdot \frac{x}{\log x}.
\end{aligned}$$

Gathering (3.1), (3.2), (3.3), (3.8) and (3.9), we get

$$T(x) \sim \left(\frac{5}{24} + \frac{1}{2} + \frac{3}{4} \right) \frac{x}{\log x} = \frac{35}{24} \frac{x}{\log x},$$

which ends the proof of Theorem 1.

IV. USE OF EXPONENTIAL SUMS.

The aim of this paragraph is to weaken the conditions (2.6) over A, A', B, B' down to the conditions appearing in Theorem 1. We improve the use of the formula

$$\sum_{(u,v) \in \mathcal{E}_p} \sum_{\substack{a \equiv u \pmod{p} \\ |a| \leq A}} \sum_{\substack{b \equiv v \pmod{p} \\ |b| \leq B}} 1 + O\left(\left(\frac{A}{p^4} + 1\right)\left(\frac{B}{p^6} + 1\right)\right)$$

giving the number of $E_{a,b}$ with $|a| \leq A, |b| \leq B$ for which a given prime $p \geq 5$ is supersingular. The idea, already appearing in [F–M], is to say that $\mathcal{E}_p$ is not too chaotic, (i. e. if $E_{a,b}$ belongs to $\mathcal{E}_p$, then E_{au^4, bu^6} is also in $\mathcal{E}_p$); it is now possible to appeal to the theory of exponential sums to detect the conditions $|au^4| \leq A, |bu^6| \leq B$. We extract from [F–M], paragraph 7 the following lemma:

LEMMA 5 *Let $0 < \alpha, \beta < p$, such that the given prime $p \geq 5$ is supersingular for $E_{\alpha, \beta}$. then the number of $E_{a,b}$, ($|a| \leq A, |b| \leq B$) isomorphic to $E_{\alpha, \beta}$ with $p^4 \nmid a$ and $p^6 \nmid b$ is*

$$\frac{4AB}{p^2} \cdot \frac{p-1}{2} + O\left((\sqrt{p} \log^2 p) \left(1 + \frac{A}{p} + \frac{B}{p}\right)\right).$$

Let us denote by $\mathcal{F}_p$ a subset of representative classes of isomorphism of the form $E_{\alpha, \beta}$ with $p \nmid \alpha\beta$. Note that the cardinality of that subset is $H(-4p) - O(1)$. By inserting the result of Lemma 5 in the formula (2.3), we now obtain

$$S(A, B, A', B') = \sum_{p \leq x} \left(|\mathcal{F}_p| \left(\frac{4AB}{p^2} \cdot \frac{p-1}{2} + O(\sqrt{p} \log^2 p (1 + \frac{A}{p} + \frac{B}{p})) \right) + O\left(\frac{AB}{p} + A + B\right) \right) \times$$

$$\left(|\mathcal{F}_p|\left(\frac{4A'B'}{p^2} \cdot \frac{p-1}{2} + O(\sqrt{p} \log^2 p (1 + \frac{A'}{p} + \frac{B'}{p}))\right) + O(\frac{A'B'}{p} + A' + B')\right),$$

where the error terms come from the curves $E_{a,b}$ and $E_{a',b'}$ with $p|ab$ and $p|a'b'$. Using now the relation $|\mathcal{F}_p| = O(\sqrt{p} \log p)$, we get

$$\begin{aligned} S(A, B, A', B') &= \sum_{p \leq x} \left(|\mathcal{F}_p| \frac{4AB}{p^2} \cdot \frac{p-1}{2} + O(p \log^3 p (\frac{A}{p} + 1)(\frac{B}{p} + 1)) \right) \times \\ &\quad \left(|\mathcal{F}_p| \frac{4A'B'}{p^2} \cdot \frac{p-1}{2} + O(p \log^3 p (\frac{A'}{p} + 1)(\frac{B'}{p} + 1)) \right) \\ &= 16ABA'B' \left(\sum_{p \leq x} \frac{|\mathcal{F}_p|^2}{p^2} \right) + O(ABA'B'), \end{aligned}$$

under the conditions $A, A' \geq x^{\frac{1}{2}+\varepsilon}$, $B, B' \geq x^{\frac{1}{2}+\varepsilon}$, $AB, A'B' \geq x^{\frac{3}{2}+\varepsilon}$. Since we have $|\mathcal{F}_p|^2 = H^2(-4p) + O(\sqrt{p} \log p)$, the treatment of the main term is straightforward by the Proposition.

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Étienne FOUVRY
Mathématique- Bâtiment 425
Université de Paris–Sud
F-91405 ORSAY Cedex

Ram MURTY
Department of Mathematics
Mc GILL University
MONTREAL, PQ
CANADA H3A 2K6